Do Now · retrieval

Floating-point representation and normalisation

Name: _____ Class: _____ Date: _____

Answer four retrieval questions before reviewing earlier learning.

Q1

Write −2 in 8-bit two's complement.

Q2

What is unsigned 00100000 after a logical right shift by one?

Q3

Convert binary 0.1_2 to decimal.

Q4

Calculate 0.5×2^3.

New Learning · key ideas

The important knowledge from today's lesson. Read, annotate, and explain it.

Core idea 1

Why is 0.0110 not normalised? — For a positive two's-complement fractional mantissa, a normalised value begins with 01, not 00.

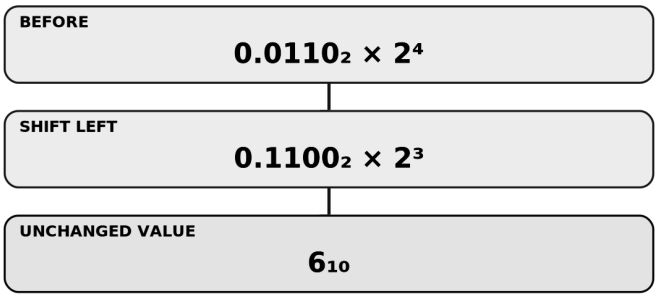
Core idea 2

Shift and compensate — Shift 0.0110 left one place to 0.1100. Decrease the exponent from 4 to 3 to compensate.

Core idea 3

Verify the value — 0.0110_2 is 0.375; 0.375 × 16 = 6. 0.1100_2 is 0.75; 0.75 × 8 = 6.

Key visual · complete concept



Annotate the key visual

Explain which labels or features show the main process.

New Learning · applying the ideas

Vocabulary · use the correct technical terms

mantissa — The signed significant part of a floating-point number.

exponent — The power of two scaling the mantissa.

normalised — A two's-complement fractional mantissa with different first two bits.

precision — Number of significant bits represented.

Check your understanding

Explain one key process in your own words using two of the terms above.

More animated examples are available in the original online lesson.

I Do · worked example

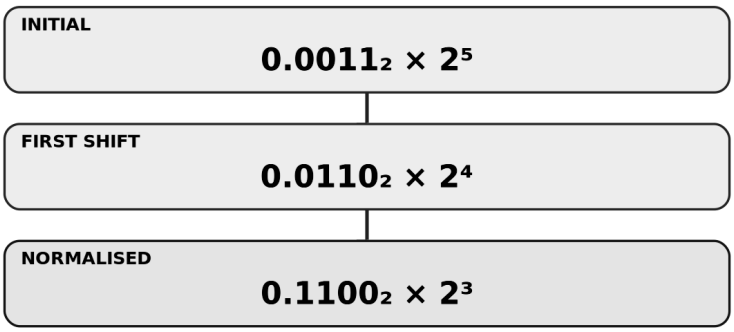
Problem

Normalise positive $0.0011_2 \times 2^5$.

Follow the worked method

1. First shift — 0.0011 becomes 0.0110; reduce exponent 5 → 4. It still begins 00.
2. Second shift — 0.0110 becomes 0.1100; reduce exponent 4 → 3. The first two bits are now 01.
3. Check — Both represent 6 in decimal, so normalisation preserved the value.

Source model · final illustrated step



Why this method works

Identify the decision, rule or invariant that makes the method correct.

We Do · Try → Discuss → Improve

TRY independently in BLACK PEN; compare methods with a partner; improve your reasoning.

Guided problem · TRY

Predict the shift
Shift the mantissa left one position: 0.0101 → 0.1010.

DISCUSS

Compare how you reached the answer. Where could someone make an error?

IMPROVE

Add one precise correction or a stronger reason. Answers are not printed here.

You Do · independent practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 1

Identify mantissa and exponent in $0.1100_2 \times 2^3$.

Question 2

Normalise positive $0.0010_2 \times 2^5$.

Check that you have used the technical terms and explained your decisions.

You Do · extended practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 3

Why can normalisation improve precision without increasing the number of mantissa bits?

Check that you have used the technical terms and explained your decisions.

Exit Ticket · reflect and retrieve

Exit Ticket

Shifting 0.0110×2^4 left to 0.1100×2^4 preserves its value.

Use key vocabulary from memory

1. Explain mantissa in your own words.

2. Explain exponent in your own words.

3. Explain normalised in your own words.

Confidence and next step

How confident are you? Thumbs up / side / down

Which question would you improve after checking the classroom model?

Check answers after the lesson using the teacher-guided source models.